Data Analysis using R

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R Markdown

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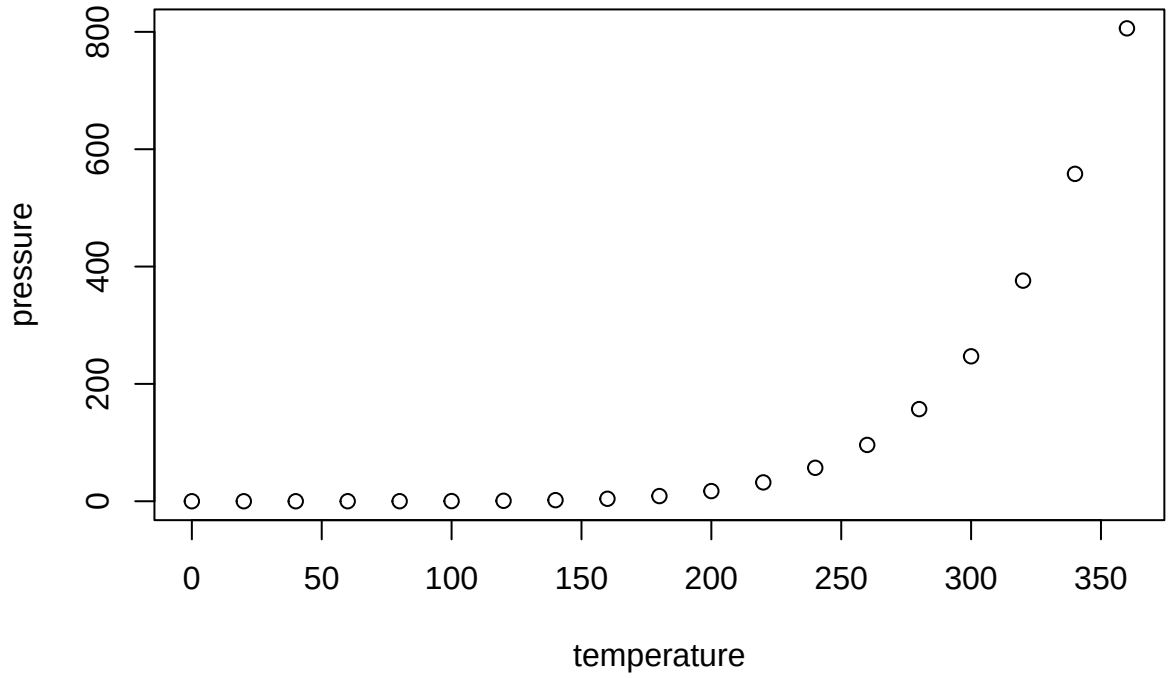
When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
summary(cars)
```

```
##      speed           dist       
##  Min.   : 4.0   Min.   :  2.00  
##  1st Qu.:12.0   1st Qu.: 26.00  
##  Median :15.0   Median : 36.00  
##  Mean   :15.4   Mean   : 42.98  
##  3rd Qu.:19.0   3rd Qu.: 56.00  
##  Max.   :25.0   Max.   :120.00
```

Including Plots

You can also embed plots, for example:



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.

```r
# Assignment 1 - Data Analysis using R Programming

# Import dataset
healthcare_dataset <- read.csv("healthcare_dataset.csv")

# View first few rows
head(healthcare_dataset)
```

```
##            Name Age Gender Blood.Type Medical.Condition Date.of.Admission
## 1 Bobby JacksOn  30   Male         B-            Cancer        2024-01-31
## 2  LesLie TErRy  62   Male         A+           Obesity        2019-08-20
## 3   DaNnY sMitH  76 Female         A-           Obesity        2022-09-22
## 4  andrEw waTtS  28 Female         O+          Diabetes        2020-11-18
## 5 adrIENNE bEll  43 Female        AB+            Cancer        2022-09-19
## 6 EMILY JOHNSOn  36   Male         A+            Asthma        2023-12-20
##             Doctor                  Hospital Insurance.Provider Billing.Amount
## 1    Matthew Smith           Sons and Miller         Blue Cross       18856.28
## 2  Samantha Davies                   Kim Inc           Medicare       33643.33
## 3 Tiffany Mitchell                  Cook PLC              Aetna       27955.10
## 4      Kevin Wells Hernandez Rogers and Vang,           Medicare       37909.78
## 5   Kathleen Hanna               White-White              Aetna       14238.32
## 6    Taylor Newton            Nunez-Humphrey   UnitedHealthcare       48145.11
##   Room.Number Admission.Type Discharge.Date  Medication Test.Results
```

```
## 1         328         Urgent     2024-02-02 Paracetamol       Normal
## 2         265      Emergency     2019-08-26   Ibuprofen Inconclusive
## 3         205      Emergency     2022-10-07     Aspirin       Normal
## 4         450       Elective     2020-12-18   Ibuprofen     Abnormal
## 5         458         Urgent     2022-10-09  Penicillin     Abnormal
## 6         389         Urgent     2023-12-24   Ibuprofen       Normal
```

```
# 1. Print structure of Dataset
str(healthcare_dataset)
```

```
## 'data.frame':    55500 obs. of  15 variables:
##  $ Name              : chr  "Bobby JacksOn" "LesLie TErRy" "DaNnY sMitH" "andrEw waTtS" ...
##  $ Age               : int  30 62 76 28 43 36 21 20 82 58 ...
##  $ Gender            : chr  "Male" "Male" "Female" "Female" ...
##  $ Blood.Type        : chr  "B-" "A+" "A-" "O+" ...
##  $ Medical.Condition : chr  "Cancer" "Obesity" "Obesity" "Diabetes" ...
##  $ Date.of.Admission : chr  "2024-01-31" "2019-08-20" "2022-09-22" "2020-11-18" ...
##  $ Doctor            : chr  "Matthew Smith" "Samantha Davies" "Tiffany Mitchell" "Kevin Wells" ...
##  $ Hospital          : chr  "Sons and Miller" "Kim Inc" "Cook PLC" "Hernandez Rogers and Vang," ...
##  $ Insurance.Provider: chr  "Blue Cross" "Medicare" "Aetna" "Medicare" ...
##  $ Billing.Amount    : num  18856 33643 27955 37910 14238 ...
##  $ Room.Number       : int  328 265 205 450 458 389 389 277 316 249 ...
##  $ Admission.Type    : chr  "Urgent" "Emergency" "Emergency" "Elective" ...
##  $ Discharge.Date    : chr  "2024-02-02" "2019-08-26" "2022-10-07" "2020-12-18" ...
##  $ Medication        : chr  "Paracetamol" "Ibuprofen" "Aspirin" "Ibuprofen" ...
##  $ Test.Results      : chr  "Normal" "Inconclusive" "Normal" "Abnormal" ...
```

```
# 2. List of variables in Dataset
names(healthcare_dataset)
```

```
##  [1] "Name"               "Age"                "Gender"
##  [4] "Blood.Type"         "Medical.Condition"  "Date.of.Admission"
##  [7] "Doctor"             "Hospital"           "Insurance.Provider"
## [10] "Billing.Amount"     "Room.Number"        "Admission.Type"
## [13] "Discharge.Date"     "Medication"         "Test.Results"
```

```
# 3. Print top 15 rows of Dataset
head(healthcare_dataset, 15)
```

```
##                 Name Age Gender Blood.Type Medical.Condition Date.of.Admission
## 1       Bobby JacksOn  30   Male         B-            Cancer        2024-01-31
## 2        LesLie TErRy  62   Male         A+           Obesity        2019-08-20
## 3         DaNnY sMitH  76 Female         A-           Obesity        2022-09-22
## 4        andrEw waTtS  28 Female         O+          Diabetes        2020-11-18
## 5       adrIENNE bEll  43 Female        AB+            Cancer        2022-09-19
## 6       EMILY JOHNSOn  36   Male         A+            Asthma        2023-12-20
## 7      edwArD EDWaRDs  21 Female        AB-          Diabetes        2020-11-03
## 8  CHrisTInA MARtinez  20 Female         A+            Cancer        2021-12-28
## 9     JASmINe aGuIlaR  82   Male        AB+            Asthma        2020-07-01
## 10   ChRISTopher BerG  58 Female        AB-            Cancer        2021-05-23
## 11   mIchElLe daniELs  72   Male         O+            Cancer        2020-04-19
## 12     aaRon MARtiNeZ  38 Female         A-      Hypertension        2023-08-13
```

```
## 13      connOR HANsEn  75 Female         A+          Diabetes        2019-12-12
## 14       rObeRt bAuer  68 Female        AB+            Asthma        2020-05-22
## 15       bROOkE brady  44 Female        AB+            Cancer        2021-10-08
##              Doctor                    Hospital Insurance.Provider
## 1     Matthew Smith             Sons and Miller         Blue Cross
## 2   Samantha Davies                      Kim Inc           Medicare
## 3  Tiffany Mitchell                    Cook PLC              Aetna
## 4       Kevin Wells  Hernandez Rogers and Vang,           Medicare
## 5    Kathleen Hanna                 White-White              Aetna
## 6     Taylor Newton              Nunez-Humphrey   UnitedHealthcare
## 7       Kelly Olson             Group Middleton           Medicare
## 8    Suzanne Thomas Powell Robinson and Valdez,              Cigna
## 9   Daniel Ferguson               Sons Rich and              Cigna
## 10      Heather Day              Padilla-Walker   UnitedHealthcare
## 11      John Duncan             Schaefer-Porter           Medicare
## 12     Douglas Mayo                 Lyons-Blair           Medicare
## 13 Kenneth Fletcher   Powers Miller, and Flores              Cigna
## 14  Theresa Freeman            Rivera-Gutierrez   UnitedHealthcare
## 15  Roberta Stewart             Morris-Arellano   UnitedHealthcare
##    Billing.Amount Room.Number Admission.Type Discharge.Date  Medication
## 1       18856.281         328         Urgent     2024-02-02 Paracetamol
## 2       33643.327         265      Emergency     2019-08-26   Ibuprofen
## 3       27955.096         205      Emergency     2022-10-07     Aspirin
## 4       37909.782         450       Elective     2020-12-18   Ibuprofen
## 5       14238.318         458         Urgent     2022-10-09  Penicillin
## 6       48145.111         389         Urgent     2023-12-24   Ibuprofen
## 7       19580.872         389      Emergency     2020-11-15 Paracetamol
## 8       45820.463         277      Emergency     2022-01-07 Paracetamol
## 9       50119.223         316       Elective     2020-07-14     Aspirin
## 10      19784.631         249       Elective     2021-06-22 Paracetamol
## 11      12576.796         394         Urgent     2020-04-22 Paracetamol
## 12       7999.587         288         Urgent     2023-09-05     Lipitor
## 13      43282.283         134      Emergency     2019-12-28  Penicillin
## 14      33207.707         309         Urgent     2020-06-19     Lipitor
## 15      40701.599         182         Urgent     2021-10-13 Paracetamol
##    Test.Results
## 1        Normal
## 2  Inconclusive
## 3        Normal
## 4      Abnormal
## 5      Abnormal
## 6        Normal
## 7  Inconclusive
## 8  Inconclusive
## 9      Abnormal
## 10 Inconclusive
## 11       Normal
## 12 Inconclusive
## 13     Abnormal
## 14       Normal
## 15       Normal
```

```r
# 4. User-defined function using a variable from the dataset
age_group <- function(age)
```

```
  {
  if (age >= 65) {
    return("Senior")
  } else {
    return("Adult")
  }
}
# Test the function
age_group(70)
```

```
## [1] "Senior"
```

```
# 5. Filter rows
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
senior_patients <- healthcare_dataset %>% filter(Age >= 65)

head(senior_patients)
```

```
##               Name Age Gender Blood.Type Medical.Condition Date.of.Admission
## 1      DaNnY sMitH  76 Female         A-            Obesity        2022-09-22
## 2  JASmINe aGuIlaR  82   Male        AB+             Asthma        2020-07-01
## 3 mIchElLe daniELs  72   Male         O+             Cancer        2020-04-19
## 4    connOR HANsEn  75 Female         A+           Diabetes        2019-12-12
## 5     rObeRt bAuer  68 Female        AB+             Asthma        2020-05-22
## 6    tIMOTHY burNs  67 Female         A-             Asthma        2023-06-28
##             Doctor                   Hospital Insurance.Provider Billing.Amount
## 1 Tiffany Mitchell                  Cook PLC              Aetna    27955.09608
## 2  Daniel Ferguson             Sons Rich and              Cigna    50119.22279
## 3      John Duncan           Schaefer-Porter           Medicare    12576.79561
## 4 Kenneth Fletcher Powers Miller, and Flores              Cigna    43282.28336
## 5  Theresa Freeman          Rivera-Gutierrez   UnitedHealthcare    33207.70663
## 6     Krista Smith                 Jones LLC         Blue Cross       42.51459
##   Room.Number Admission.Type Discharge.Date  Medication Test.Results
## 1         205      Emergency     2022-10-07     Aspirin       Normal
## 2         316       Elective     2020-07-14     Aspirin     Abnormal
## 3         394         Urgent     2020-04-22 Paracetamol       Normal
## 4         134      Emergency     2019-12-28  Penicillin     Abnormal
## 5         309         Urgent     2020-06-19     Lipitor       Normal
## 6         115       Elective     2023-07-02     Aspirin       Normal
```

```r
# 6. Dependent and independent variables

# Independent variable
age_data <- healthcare_dataset["Age"]

# Dependent variable
billing_data <- healthcare_dataset["Billing.Amount"]

# Combine both variables
healthcare_analysis <- cbind(age_data, billing_data)
# Display the new data frame
head(healthcare_analysis)
```

```
##   Age Billing.Amount
## 1  30       18856.28
## 2  62       33643.33
## 3  76       27955.10
## 4  28       37909.78
## 5  43       14238.32
## 6  36       48145.11
```

```r
# 7. Check for missing values
sum(is.na(healthcare_dataset))
```

```
## [1] 0
```

```r
# No missing values were found, so no rows were removed

# 8. Check for duplicated rows
sum(duplicated(healthcare_dataset))
```

```
## [1] 534
```

```r
# 8A  Extra step -Remove duplicated rows
healthcare_clean <- healthcare_dataset %>% distinct()
nrow(healthcare_clean)
```

```
## [1] 54966
```

```r
# 9. Reorder rows by Billing Amount in descending order
healthcare_sorted <- healthcare_clean %>% arrange(desc(Billing.Amount))
head(healthcare_sorted, 10)
```

```
##                Name Age Gender Blood.Type Medical.Condition Date.of.Admission
## 1     tOdd CARrILlO  51 Female         A+      Hypertension        2023-09-08
## 2       kARen klInE  79 Female        AB+            Cancer        2021-06-19
## 3       kARen klInE  74 Female        AB+            Cancer        2021-06-19
## 4    DAVId SanDOvaL  22   Male        AB-      Hypertension        2019-11-02
## 5  kathRYN GoNzales  18 Female         O+          Diabetes        2020-08-11
## 6    BRETT mARshAll  42 Female         B+            Asthma        2021-11-10
```

```
## 7       LAuRie hOod  42 Female         A-         Arthritis        2021-06-27
## 8       LAuRie hOod  43 Female         A-         Arthritis        2021-06-27
## 9      JUstIn CLark  53 Female        AB+            Cancer        2019-05-30
## 10     sCotT pOWElL  38 Female         B+            Cancer        2019-05-20
##               Doctor         Hospital Insurance.Provider Billing.Amount
## 1   Kathleen Griffin    Griffin Group         Blue Cross       52764.28
## 2  Dr. Joseph Gordon Hernandez-Morton   UnitedHealthcare       52373.03
## 3  Dr. Joseph Gordon Hernandez-Morton   UnitedHealthcare       52373.03
## 4       Taylor Novak  Sons and Bailey         Blue Cross       52271.66
## 5      Kristin Clark       Miller Ltd              Aetna       52211.85
## 6     Robert Hartman       PLC Garner              Aetna       52181.84
## 7        Brian Riley    Walker-Garcia              Cigna       52170.04
## 8        Brian Riley    Walker-Garcia              Cigna       52170.04
## 9       Andrea Jones     Ruiz-Anthony   UnitedHealthcare       52154.24
## 10  Stephen Hood PhD  George-Gonzalez              Cigna       52102.24
##    Room.Number Admission.Type Discharge.Date  Medication Test.Results
## 1          209       Elective     2023-10-04   Ibuprofen       Normal
## 2          181         Urgent     2021-07-03  Penicillin Inconclusive
## 3          181         Urgent     2021-07-03  Penicillin Inconclusive
## 4          228      Emergency     2019-11-11   Ibuprofen       Normal
## 5          383         Urgent     2020-08-12 Paracetamol Inconclusive
## 6          140      Emergency     2021-11-17  Penicillin Inconclusive
## 7          306         Urgent     2021-06-29  Penicillin       Normal
## 8          306         Urgent     2021-06-29  Penicillin       Normal
## 9          399         Urgent     2019-06-22     Aspirin       Normal
## 10         155      Emergency     2019-05-29   Ibuprofen       Normal
```

```r
# 10. Rename columns
healthcare_renamed <- healthcare_clean %>% rename(
Billing_Amount = Billing.Amount,
Medical_Condition = Medical.Condition
)
names(healthcare_renamed)
```

```
##  [1] "Name"               "Age"                "Gender"
##  [4] "Blood.Type"         "Medical_Condition"  "Date.of.Admission"
##  [7] "Doctor"             "Hospital"           "Insurance.Provider"
## [10] "Billing_Amount"     "Room.Number"        "Admission.Type"
## [13] "Discharge.Date"     "Medication"         "Test.Results"
```

```r
# 11. Create a new variable using a mathematical function
healthcare_new <- healthcare_renamed %>% mutate(Billing_Double = Billing_Amount * 2)
head(healthcare_new)
```

```
##           Name Age Gender Blood.Type Medical_Condition Date.of.Admission
## 1 Bobby JacksOn  30   Male         B-            Cancer        2024-01-31
## 2  LesLie TErRy  62   Male         A+           Obesity        2019-08-20
## 3   DaNnY sMitH  76 Female         A-           Obesity        2022-09-22
## 4  andrEw waTtS  28 Female         O+          Diabetes        2020-11-18
## 5 adrIENNE bEll  43 Female        AB+            Cancer        2022-09-19
## 6 EMILY JOHNSOn  36   Male         A+            Asthma        2023-12-20
##            Doctor                  Hospital Insurance.Provider Billing_Amount
## 1   Matthew Smith           Sons and Miller         Blue Cross       18856.28
```

```
## 2  Samantha Davies                    Kim Inc           Medicare       33643.33
## 3 Tiffany Mitchell                   Cook PLC              Aetna       27955.10
## 4      Kevin Wells Hernandez Rogers and Vang,           Medicare       37909.78
## 5   Kathleen Hanna                White-White              Aetna       14238.32
## 6    Taylor Newton             Nunez-Humphrey   UnitedHealthcare       48145.11
##   Room.Number Admission.Type Discharge.Date  Medication Test.Results
## 1         328         Urgent     2024-02-02 Paracetamol       Normal
## 2         265      Emergency     2019-08-26   Ibuprofen Inconclusive
## 3         205      Emergency     2022-10-07     Aspirin       Normal
## 4         450       Elective     2020-12-18   Ibuprofen     Abnormal
## 5         458         Urgent     2022-10-09  Penicillin     Abnormal
## 6         389         Urgent     2023-12-24   Ibuprofen       Normal
##   Billing_Double
## 1       37712.56
## 2       67286.65
## 3       55910.19
## 4       75819.56
## 5       28476.64
## 6       96290.22
```

```r
# 12. Create a training set using random sampling

set.seed(123)

training_data <- healthcare_new %>% sample_frac(0.80)
nrow(training_data)
```

```
## [1] 43973
```

```r
# 13. Summary statistics
summary(healthcare_new)
```

```
##         Name                Age              Gender          Blood.Type
##  Length   :54966   Min.   :13.00   Length   :54966   Length   :54966
##  N.unique :49992   1st Qu.:35.00   N.unique :    2   N.unique :    8
##  N.blank  :    0   Median :52.00   N.blank  :    0   N.blank  :    0
##  Min.nchar:    6   Mean   :51.54   Min.nchar:    4   Min.nchar:    2
##  Max.nchar:   28   3rd Qu.:68.00   Max.nchar:    6   Max.nchar:    3
##                    Max.   :89.00
##  Medical_Condition Date.of.Admission       Doctor           Hospital
##  Length   :54966   Length   :54966   Length   :54966   Length   :54966
##  N.unique :    6   N.unique : 1827   N.unique :40341   N.unique :39876
##  N.blank  :    0   N.blank  :    0   N.blank  :    0   N.blank  :    0
##  Min.nchar:    6   Min.nchar:   10   Min.nchar:    6   Min.nchar:    6
##  Max.nchar:   12   Max.nchar:   10   Max.nchar:   28   Max.nchar:   35
##
##  Insurance.Provider Billing_Amount   Room.Number      Admission.Type
##  Length   :54966    Min.   :-2008   Min.   :101.0   Length   :54966
##  N.unique :    5    1st Qu.:13244   1st Qu.:202.0   N.unique :    3
##  N.blank  :    0    Median :25543   Median :302.0   N.blank  :    0
##  Min.nchar:    5    Mean   :25544   Mean   :301.1   Min.nchar:    6
##  Max.nchar:   16    3rd Qu.:37820   3rd Qu.:401.0   Max.nchar:    9
##                     Max.   :52764   Max.   :500.0
```

```
##    Discharge.Date      Medication       Test.Results   Billing_Double
##  Length   :54966   Length   :54966   Length   :54966   Min.   : -4017
##  N.unique : 1856   N.unique :    5   N.unique :    3   1st Qu.: 26487
##  N.blank  :    0   N.blank  :    0   N.blank  :    0   Median : 51085
##  Min.nchar:   10   Min.nchar:    7   Min.nchar:    6   Mean   : 51089
##  Max.nchar:   10   Max.nchar:   11   Max.nchar:   12   3rd Qu.: 75640
##                                                       Max.   :105529
```

```r
# 14. Mean of Age
mean(healthcare_new$Age)
```

```
## [1] 51.53519
```

```r
# 15. Median of Age
median(healthcare_new$Age)
```

```
## [1] 52
```

```r
# 16. Mode of Age
get_mode <- function(x) {
  ux <- unique(x)
  ux[which.max(tabulate(match(x, ux)))]
}

get_mode(healthcare_new$Age)
```

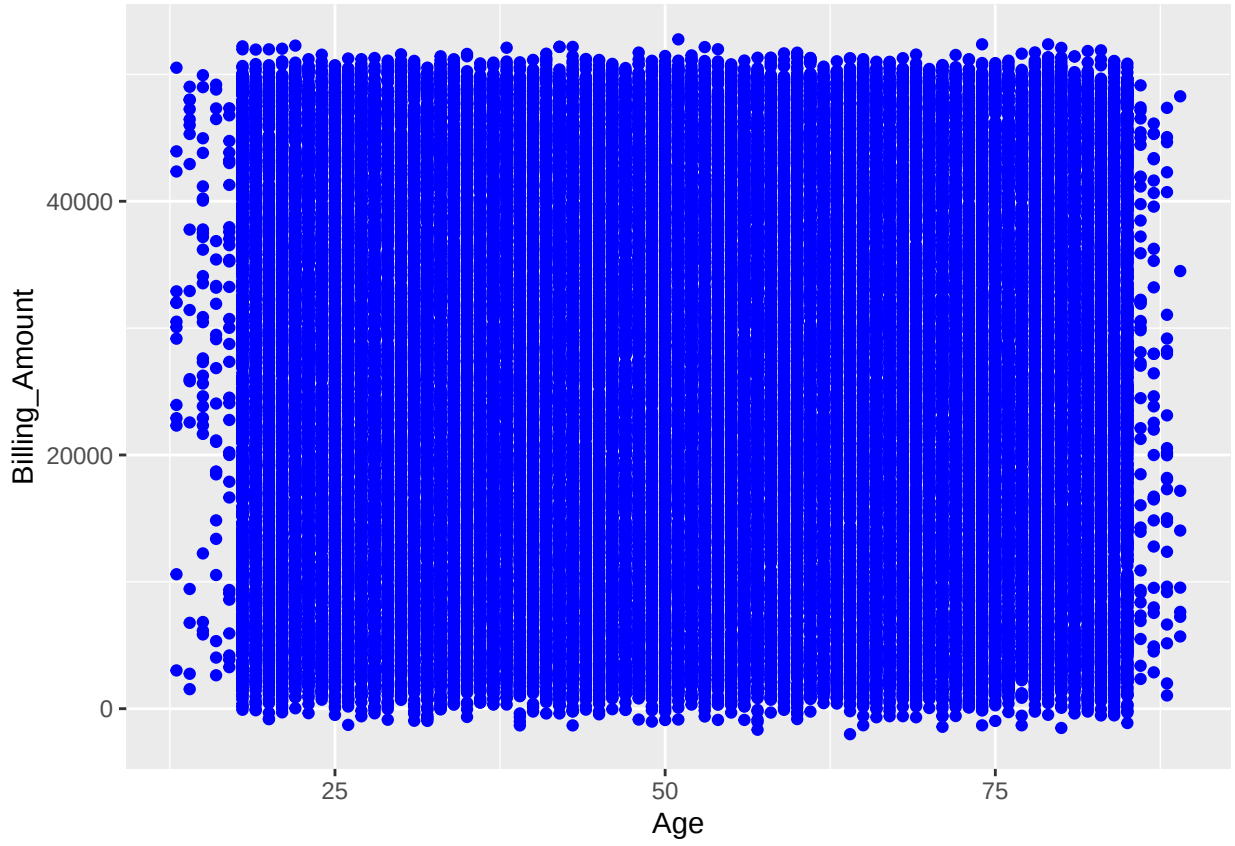
```
## [1] 38
```

```r
# 17. Range of Age
range(healthcare_new$Age)
```

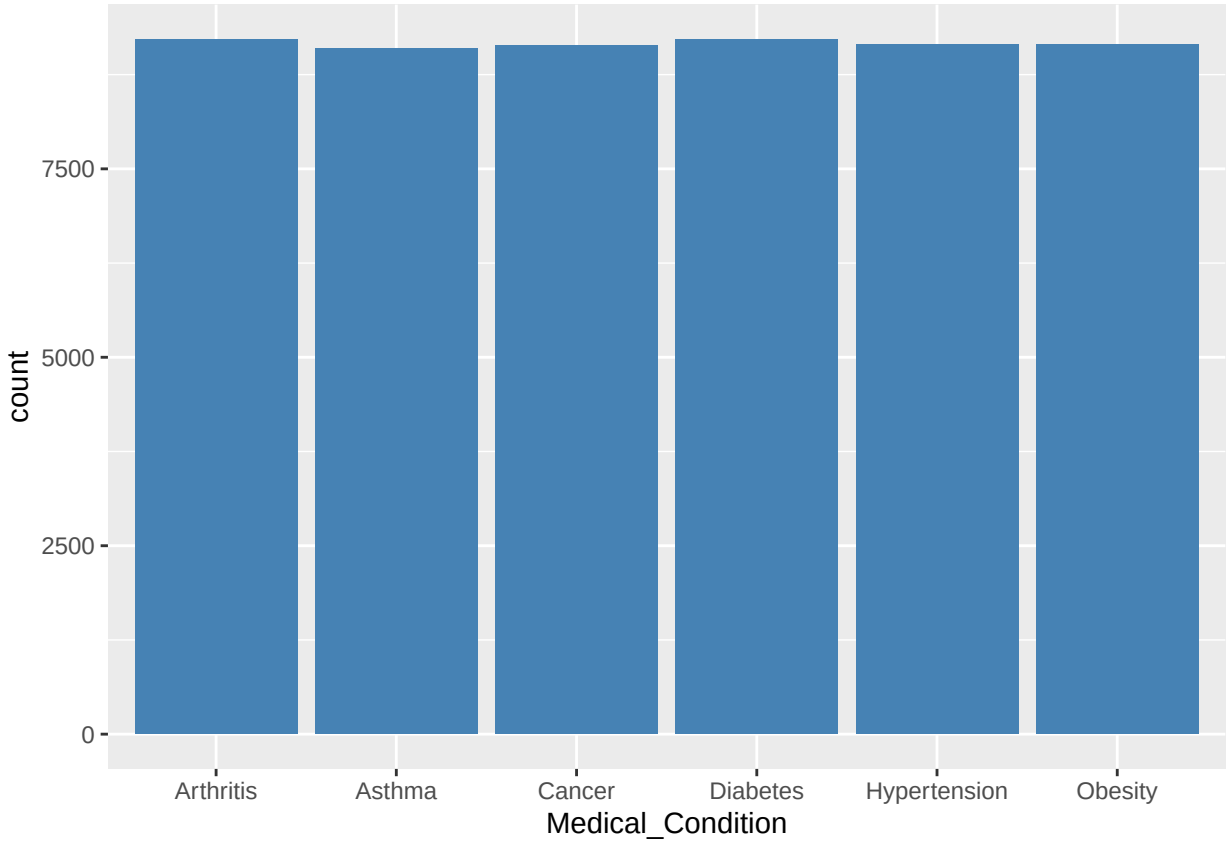
```
## [1] 13 89
```

```r
# 18. Scatter plot
library(ggplot2)

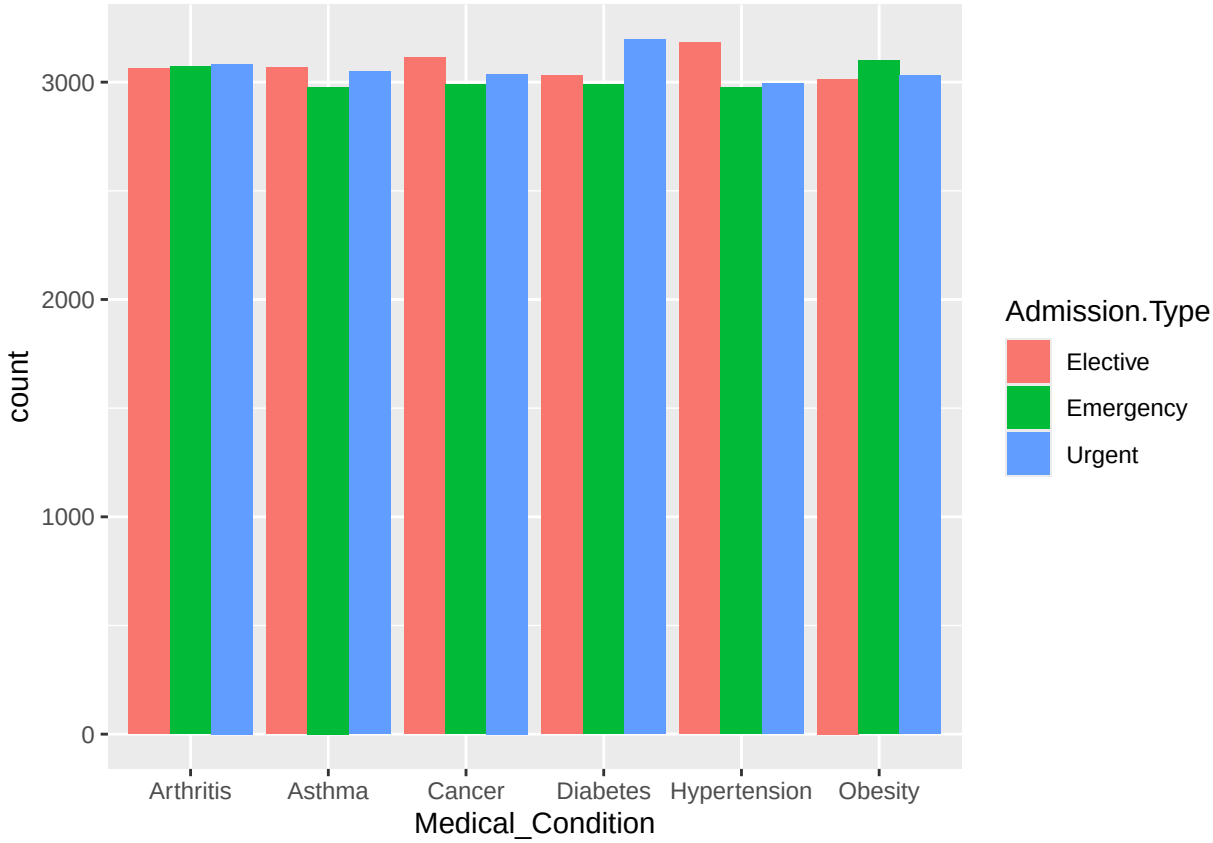
ggplot(healthcare_new, aes(x = Age, y = Billing_Amount)) + geom_point(color= "blue")
```




```
# 19. Bar plot (1 variable)
ggplot(healthcare_new, aes(x = Medical_Condition)) + geom_bar(fill = "steelblue")
```




```
# 19A. Bar plot using two variables
ggplot(healthcare_new, aes(x = Medical_Condition, fill = Admission.Type)) +
  geom_bar(position = "dodge")
```




```
# 20. Pearson correlation
cor(healthcare_new$Age, healthcare_new$Billing_Amount, method = "pearson")
```

```
## [1] -0.0034267
```